LAB 01

DEPLOY YOUR FIRST STATIC WEBSITE TO THE CLOUD

Lecturer: Truong Thi Ngoc Phuong

Course	Introduction to Cloud Computing
Duration / Type	90 minutes · Individual assignment
Score / Submission	10 points · HTTPS URL + Source Code + PDF Report

1. Lab Overview

Create a static website and deploy it to a cloud-hosted platform. Your website must be publicly accessible through HTTPS. Choose ONE of the following options; all are graded equally.

Option	Hosting platform	Account requirement
A	AWS Amplify Hosting	Eligible AWS access; verify service availability
B	GitHub Pages	Free GitHub account; public repository
C	Cloudflare Pages	Free Cloudflare account

Students are not required to purchase an AWS account or provide a payment card to complete this lab. If AWS access is unavailable, choose B or C.

2. Learning Objectives

1. Explain cloud-based static website hosting.
2. Create a basic HTML/CSS website.
3. Deploy and verify a public HTTPS URL.
4. Compare local files with cloud-hosted websites.
5. Document deployment evidence and results.

3. Lab Tasks

Task 1 — Create a Static Website

Website title: My First Cloud Website

- Student full name, student ID and course name.
- Navigation menu with at least three sections.
- About Me section.
- Cloud Computing section (at least 80 words in your own words).
- Cloud Services section: explain IaaS, PaaS and SaaS with one example each.
- Deployment Information section identifying the chosen platform.
- At least one locally stored illustration/image, custom CSS, and a responsive desktop/mobile layout.

Minimum files: index.html, style.css, and assets/ with at least one image. All navigation links, images and CSS must work. No backend or database is required.

Task 2 — Deploy the Website

Option A — AWS Amplify Hosting

1. Sign in to an eligible AWS account or permitted AWS Academy environment.
2. Open AWS Amplify Hosting and create a deployment using an available method.
3. Upload or connect your website files.
4. Wait for successful deployment, then open and record the HTTPS URL.

Official tutorial: https://docs.aws.amazon.com/hands-on/latest/host-static-website/host-static-website.html

Note: AWS Academy Learner Lab may restrict Amplify. Check with your instructor first. Do not create billable resources without permission.

Option B — GitHub Pages

1. Create a public repository named cloud-lab1.
2. Upload website files to the main branch.
3. Go to Settings → Pages → Deploy from a branch.
4. Choose main and / (root), then Save.
5. Open the published HTTPS URL.

Official tutorial: https://docs.github.com/en/pages/quickstart

Option C — Cloudflare Pages

1. Sign in to Cloudflare and open Workers & Pages.
2. Create a Pages project using Direct Upload.
3. Upload the website folder or ZIP.
4. Wait for deployment and open the pages.dev HTTPS URL.

Official tutorial: https://developers.cloudflare.com/pages/get-started/direct-upload/

Task 3 — Verify Deployment

- Public HTTPS URL opens in a browser and an incognito window without signing in.
- All text, CSS and images load correctly.
- Navigation works and layout is usable on desktop and mobile.

Task 4 — Explain Your Deployment

Answer each question in approximately 50–100 words:

1. Which platform did you select, and why?
2. How does opening index.html locally differ from visiting the deployed HTTPS URL?
3. What is HTTPS, and why is it important?
4. Does the selected hosting platform fit IaaS, PaaS or SaaS directly? Explain its characteristics.
5. What changes would be needed for authentication or a database?

4. Required Submission

Upload ONE ZIP named Lab01_StudentID_FullName.zip and paste your HTTPS URL separately into the LMS submission field.

ZIP structure (example):

```
Lab01_23520001_NguyenVanA.zip
```

```
index.html
style.css
assets/cloud.svg
Lab01_Report_23520001.pdf
```

5. PDF Report — Required Evidence

Evidence	What must be visible
E1 — Desktop website	Full website with student name and ID
E2 — Mobile website	Responsive view in a narrow/mobile viewport
E3 — Hosting dashboard	Platform/project name and successful deployment or Pages settings
E4 — HTTPS verification	Browser address bar showing the public https:// URL
E5 — Written answers	Complete answers to Q1–Q5

All screenshots must be from your own deployment. Do not include passwords, API keys or private credentials. Keep the URL accessible until grading is complete.

6. Illustrative Student Submission

Example only — replace all placeholder data and URLs with your own deployment evidence.

Field	Example
Student	Nguyen Van A — 23520001
Selected option	B — GitHub Pages
Repository	github.com/USERNAME/cloud-lab1
Public URL	https://USERNAME.github.io/cloud-lab1/
Website title	My First Cloud Website

Example website layout

Header: “My First Cloud Website” · navigation: About | Cloud Services | Deployment

About Me: student name and ID · Cloud Computing: 80+ word explanation · Services: IaaS/PaaS/SaaS cards · Deployment: platform name and HTTPS link.

7. Grading Rubric

Criterion	Points
Required website content and sections	2.0
HTML/CSS, navigation and responsive design	2.0
Successful public HTTPS deployment	2.5
Deployment evidence and screenshots	1.5
Five conceptual answers	2.0
TOTAL	10.0

8. Submission Checklist

- Required sections and student information are correct.
- Public HTTPS URL works without login.

- CSS, images, navigation and mobile layout work.
- Four deployment screenshots and five written answers are included.
- ZIP and PDF filenames follow the required convention.
- HTTPS URL is also submitted separately in the LMS.